

Simple Event Volatility Model

Linus Too

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Consider the discrete-time stochastic volatility process given by the asset price S_i and its volatility σ_i at time t_i :

$$S_{i+1} = S_i + \sigma_i S_i \epsilon_i, \quad (1)$$

$$\begin{aligned} \sigma_{i+1}^n &= \omega_{i+1}^{n/2} \times \left[\sigma_i^n + \alpha_i \Delta t_i + \nu_i (\rho_i \epsilon_i + \sqrt{1 - \rho_i^2} \eta_i) \right], \\ &= \omega_{i+1}^{n/2} \times F_i(\sigma_i, S_i), \end{aligned} \quad (2)$$

where ϵ_i and η_i are independent normal random variables sampled from $\mathcal{N}(0, \Delta t_i)$, Δt_i is the time increment, ω_{i+1} is a positive random variable representing event weighting that equals to one in the absence of events, α_i , ν_i , and ρ_i can be functions of t_i , S_i , and σ_i , and they respectively (loosely) represent the rate of change of vol/var, the vol of vol, and the correlation between the asset and its vol/var. Note $n = 1$ corresponds to a stochastic volatility model, while $n = 2$ corresponds to a stochastic variance model.

Now, for simplicity, we will assume the market observables at time t_i are the asset price S_i and its expected sum of log return from t_i to t_j , which is given by

$$\mathbb{E} \left[\sum_{k=i}^{j-1} \log \left(\frac{S_{k+1}}{S_k} \right) \right] = \mathbb{E} \left[\sum_{k=i}^{j-1} \log (1 + \sigma_k \epsilon_k) \right] \quad (3)$$

$$\approx \mathbb{E} \left[\sum_{k=i}^{j-1} \sigma_k \epsilon_k - \frac{1}{2} \sigma_k^2 \epsilon_k^2 \right] \quad (4)$$

$$= -\frac{1}{2} \sum_{k=i}^{j-1} \mathbb{E} [\sigma_k^2 \Delta t_k]. \quad (5)$$

Therefore at time t_i , we assume the instantaneous volatility σ_i is observable. But we will also make the assumption that σ_i is observed with a slight time lag $\delta t_i \ll \Delta t_i$, to avoid the instantaneous causality issue.

The σ_{i+1} process is "driven" by the event weighting ω_{i+1} and the stochastic terms ϵ_i and η_i . So given the market observable at $t < t_i + \delta t_i$, $S_{\leq i+1}, \sigma_{\leq i}$, we can predict σ_{i+1} by calculating the expected value of σ_{i+1} conditional on the information available:

$$\mathbb{E} [\sigma_{i+1}^n | S_{i+1}, S_i, \sigma_i] = \mathbb{E} \left[\omega_{i+1}^{n/2} \times \left[\sigma_i^n + \alpha_i \Delta t_i + \nu_i (\rho_i \epsilon_i + \sqrt{1 - \rho_i^2} \eta_i) \right] | S_{i+1}, S_i, \sigma_i \right] \quad (6)$$

$$= \mathbb{E} \left[\omega_{i+1}^{n/2} | S_{i+1}, S_i, \sigma_i \right] \times \left[\sigma_i^n + \alpha_i \Delta t_i + \nu_i \rho_i \frac{S_{i+1} - S_i}{\sigma_i S_i} \right]. \quad (7)$$

with variance

$$\text{Var} [\sigma_{i+1}^n \mid S_{i+1}, S_i, \sigma_i] = \text{Var} \left[\omega_{i+1}^{n/2} \times \left[\sigma_i^n + \alpha_i \Delta t_i + \nu_i (\rho_i \epsilon_i + \sqrt{1 - \rho_i^2} \eta_i) \right] \mid S_{i+1}, S_i, \sigma_i \right] \quad (8)$$

$$\begin{aligned} &= \text{Var} \left[\omega_{i+1}^{n/2} \mid S_{i+1}, S_i, \sigma_i \right] \times \left[\sigma_i^n + \alpha_i \Delta t_i + \nu_i \rho_i \frac{S_{i+1} - S_i}{\sigma_i S_i} \right]^2 \\ &\quad + \mathbb{E} [\omega_{i+1}^n \mid S_{i+1}, S_i, \sigma_i] \times \nu_i^2 (1 - \rho_i^2) \Delta t. \end{aligned} \quad (9)$$

Now the problem is given a historic past series of $S_{\leq i}, \sigma_{\leq i}$, without knowing the function/process $F_i(\sigma_i, S_i)$, how can we use causal inference to estimate the $\omega_{>i}$ or its (conditional) distribution? What is the parametric and non-parametric approach deduce the "causal structure" of events, S_i and σ_i ? How can we use the estimated $\omega_{>i}$ to predict the future volatility $\sigma_{>i}$?

Exercise:

Generate the series of $S_{\leq i}, \sigma_{\leq i}$ using the above stochastic volatility model with constant parameters and event weight for one periodic event.

- 1. Calculate the Granger causality/TDMI between the volatility and the binary event indicator.*
- 2. Deduce the causal structure using a parametric approach and a non-parametric approach.*
- 3. Estimate the event weight $\omega_{>i}$ using a parametric approach and a non-parametric approach.*
- 4. Use the estimated event weight $\omega_{>i}$ to predict the future volatility $\sigma_{>i}$.*
- 5. Analyse the prediction accuracy and the calibration of the model.*
- 6. Repeat the above steps for different values and/or functional forms of the underlying parameters.*